RRA

LAB 3 (Parteneriat cu laburile scrise de PETrascu)

Stabilizarea sistemelor MIMO + Youla pe MIMO

Stabilitate BIBO (Bounded Input Bounded Output)

Ce inseamna acest fel de stabilitate? Daca am o intrare marginita u $||u||_\infty < \infty$, atunci am un Σ care imi produce o iesire marginita y $||y||_\infty < \infty$

$y = \Sigma(u)$, unde y este raspuns fortat

Sa luam reprezentarea pe stare:

$\begin{cases} \dot{x} = \mathrm{Ax} + \mathrm{Bu} \\ y = \mathrm{Cx} + \mathrm{Du} \end{cases} \Rightarrow \begin{cases} Y(s) = G(s)U(s) \\ G(s) = C(s)(\mathrm{sI} - A(s))^{-1}B(s) + D(s) \end{cases}$. Polii lui $G(s) \in \mathbb{C}_-$. Daca sistemul este **minimal**, atunci acesti poli coincid cu $\Lambda(A)$

Stabilitate interna asimptotica

$$\begin{cases} \dot{x} = \mathrm{Ax} + \mathrm{Bu} \\ y = \mathrm{Cx} + \mathrm{Du} \end{cases}$$

Daca $u(t) \equiv 0 \Rightarrow x_l(t) = e^{\mathrm{At}}x_0$, unde x_l este raspuns liber. De asemenea,

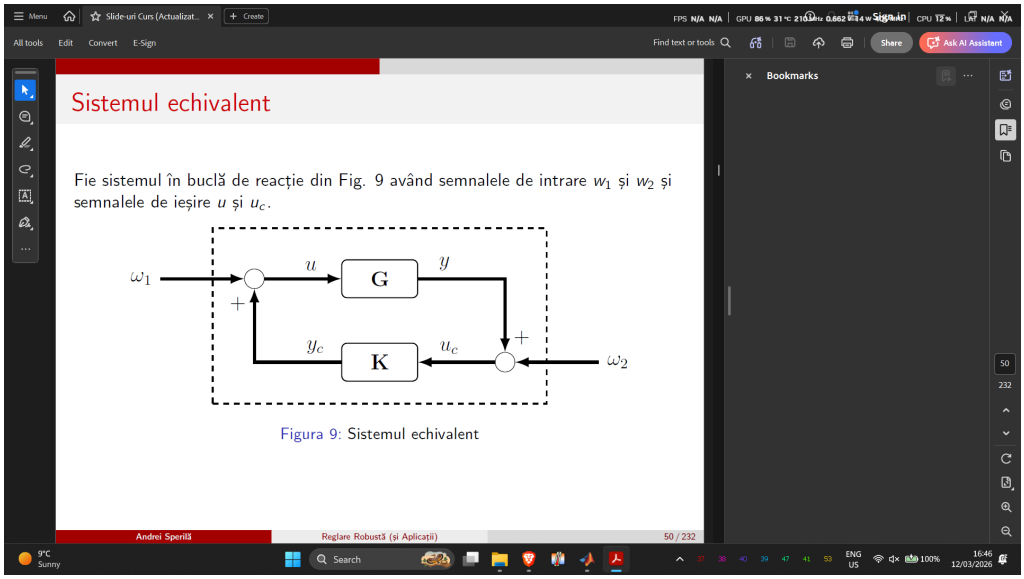
$\lim_{t\to\infty}||x_l(t)||_p = 0, \forall x_0 \in \mathbb{R}^n$ si $p \in \mathbb{N} - \{0\} \Leftrightarrow \Lambda(A) \in \mathbb{C}_-$

In acest sens, avem schema urmatoare:

$$\begin{cases} \text{Stabilitate interna} \Rightarrow \text{stabil BIBO} \\ \text{Stabilitate BIBO} + \text{controlabilitate si observabilitate} \Rightarrow \text{Stabilitate interna} \end{cases}$$

$$\begin{aligned} \text{Internal stability} &\Longrightarrow \text{BIBO stability} \\ \text{Internal stability} &\Longleftarrow \text{BIBO stability} + \\ &\qquad \text{controllability and observability} \end{aligned}$$

De ce este asa? Cand ma uit la stabilitate interna, ma uit la valorile proprii ale lui A $\Lambda(A)$care cuprind *atat polii lui* $G(s)$, cat si partile *necontrolabile* (cele unde nu poate "ajunge") si *neobservabile* (ce nu poate "vedea")



si am asa: $\begin{cases} G: \begin{cases} \dot{x} = \mathrm{Ax} + \mathrm{Bu} \\ y = \mathrm{Cx} + \mathrm{Du} \end{cases} \\ K: \begin{cases} \dot{x_k} = A_k x_k + B_k u_k \\ y_k = C_k x_k + D_k u_k \end{cases} \end{cases}$. In acest sens, avem $\begin{cases} \dot{\begin{bmatrix} x \\ x_k \end{bmatrix}} = A_{\mathrm{cl}} \begin{bmatrix} x \\ x_k \end{bmatrix} + B_{\mathrm{cl}} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \\ \begin{bmatrix} y \\ y_k \end{bmatrix} = C_{\mathrm{cl}} \begin{bmatrix} x \\ x_k \end{bmatrix} + D_{\mathrm{cl}} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \end{cases}$. Noi trebuie sa aflam $K = ?$

astfel incat $\Lambda(A_{\mathrm{cl}}) \in \mathbb{C}_-$. In acest sens $G_{\mathrm{cl}}(s) = C_{\mathrm{cl}}(s)(\mathrm{sI} - A_{\mathrm{cl}}(s))^{-1} B_{\mathrm{cl}}(s) + D_{\mathrm{cl}}(s)$ este matrice de transfer stabila $(\text{cu poli} \in \mathbb{C}_-)$

Acest sistem este bine definit in sens strict $\Leftrightarrow \forall \begin{cases} I - K(\infty)G(\infty) \\ I - G(\infty)K(\infty) \\ \begin{bmatrix} I & -K(\infty) \\ -G(\infty) & I \end{bmatrix} \end{cases}$ **sa fie inversabila. Una din cele 3 este suficient, caci sunt echivalente intre ele**

O chestie smechera

Daca G si K sunt ***<u>MINIM Stabilizabile si Detectabile,</u>*** atunci stabilizarea in bucla inchisa **este echivalenta cu** stabiizarea interna asimptotica in bucla inchisa

Remarca:

Rezolvarea stabilizarii este grea, dar echivalenta cu probleme usoare de stabilizare I/O

Acum sa trecem in Lapalce

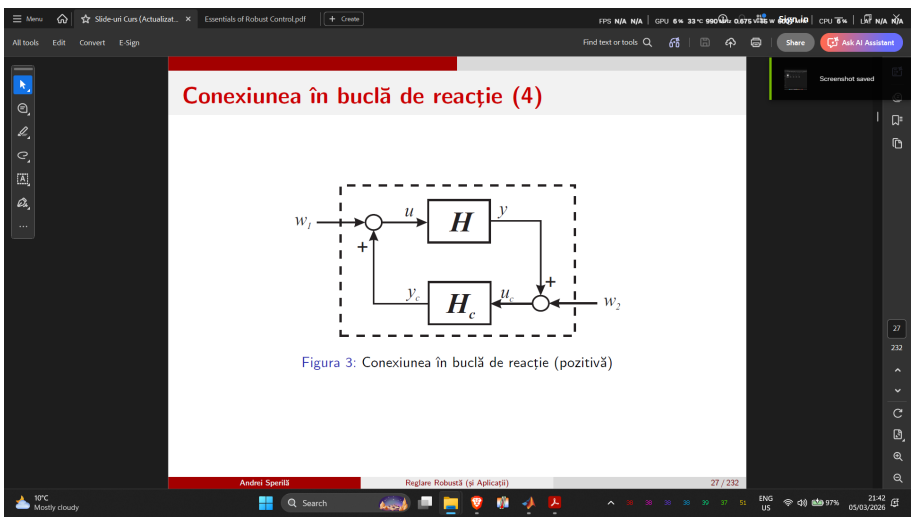
$\begin{cases} y = G(w_1 + y_k) \\ y_k = K(w_2 + y) \end{cases}$. Acum vom face 2 inlocuiri si anume $y_k \to y$ si $y \to y_k$. In acest sens, avem:

$$\begin{cases} y = \mathrm{G}w_1 + \mathrm{GK}w_2 + \mathrm{GK}y \\ y_k = \mathrm{K}w_2 + \mathrm{KG}w_1 + \mathrm{KG}y_k \end{cases} \Rightarrow \begin{cases} y(I - \mathrm{GK}) = \mathrm{G}w_1 + \mathrm{GK}w_2 \\ y_k(I - \mathrm{KG}) = \mathrm{K}w_2 + \mathrm{KG}w_1 \end{cases}.$$ Pentru buna definire, avem nevoie de $\begin{bmatrix} I & -D \\ -D_k & I \end{bmatrix}$ sa fie inversabila. Acum sa reluam calculele:

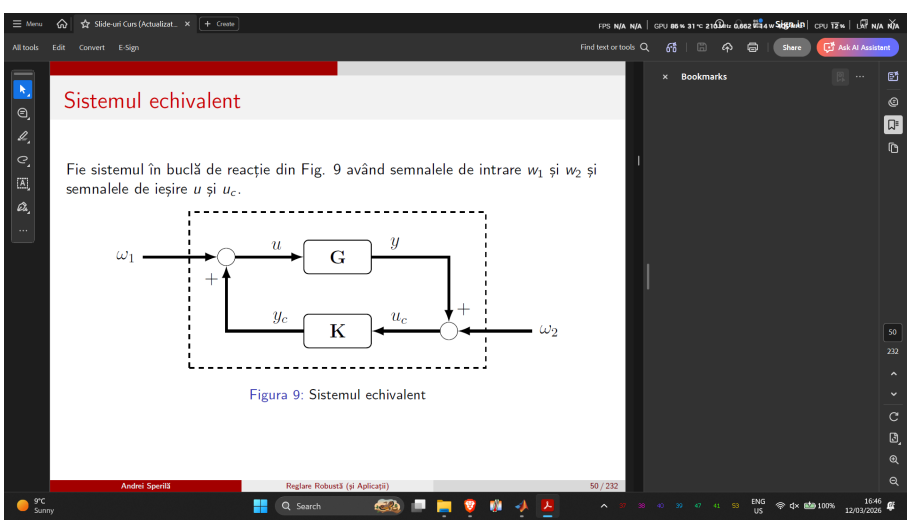
$$\begin{cases} y = (I - \mathrm{GK})^{-1}\mathrm{G}w_1 + (I - \mathrm{GK})^{-1}\mathrm{GK}w_2 \\ y_k = (I - \mathrm{KG})^{-1}\mathrm{K}w_2 + (I - \mathrm{KG})^{-1}\mathrm{KG}w_1 \end{cases} \Rightarrow \begin{cases} y = (I - \mathrm{GK})^{-1}\mathrm{G}w_1 + (I - \mathrm{GK})^{-1}\mathrm{GK}w_2 \\ y_k = K(I - \mathrm{GK})^{-1}w_1 + K(I - \mathrm{GK})^{-1}\mathrm{G}w_2 \end{cases}$$

In acest sens, vrem $\begin{cases} (I - \mathrm{GK})^{-1}G \\ (I - \mathrm{GK})^{-1}\mathrm{GK} \\ K(I - \mathrm{GK})^{-1} \\ K(I - \mathrm{GK})^{-1}G \end{cases} \Rightarrow \begin{cases} (I - \mathrm{GK})^{-1}G \\ (I - \mathrm{GK})^{-1}(\mathrm{GK} - I + I) \\ K(I - \mathrm{GK})^{-1} \\ K(I - \mathrm{GK})^{-1}G \end{cases} \Rightarrow$

$\begin{cases} (I - \mathrm{GK})^{-1}G \\ (I - \mathrm{GK})^{-1} - I \\ K(I - \mathrm{GK})^{-1} \\ K(I - \mathrm{GK})^{-1}G \end{cases}$ sa fie stabile. Mai precis, $\begin{bmatrix} K \\ I \end{bmatrix}(I - \mathrm{GK})^{-1}[G \quad I]$ sa fie stabila, cu poli in $\mathbb{C}_-$. Asta se mai numeste si **"The gang of four" (zic sa se faca film despre asta, o sa propun asta)**



stabila →



(laplace) stabila

Fie n_G si n_K **numarul de poli instabili** $(\mathbb{C}_+)$ **ai lui G respectiv K. Sistemul in bulca de reactie este intern stabil** $\Leftrightarrow$ **este bine definit in sens strict si** $\begin{cases} \text{numarul de poles}(G(s)K(s)) = n_G + n_K \\ (I - \mathrm{GK})^{-1} \text{ este in } \mathscr{RH}_\infty \end{cases}$. Prima conditie ne spune ca **nu exista simplificari poli-zero instabile** cand se formenaza produsul GK, iar a doua conditie ne spune ca $I - \mathrm{GK}$ **nu are zerouri instabile**

Atunci cand G si / sau K sunt stabile in $\mathscr{RH}_\infty$, stabilitatea interna a buclei se poate determina usor

$$\begin{cases} G \in \mathscr{RH}_\infty, \text{sistemul este intern stabil} \Leftrightarrow K(I - \mathrm{GK})^{-1} \in \mathscr{RH}_\infty \\ K \in \mathscr{RH}_\infty, \text{sistemul este intern stabil} \Leftrightarrow G(I - \mathrm{KG})^{-1} \in \mathscr{RH}_\infty \\ G \in \mathscr{RH}_\infty \text{ si } K \in \mathscr{RH}_\infty, \text{sistemul este intern stabil} \Leftrightarrow (I - \mathrm{GK})^{-1} \in \mathscr{RH}_\infty \end{cases}$$

Acum intram in prima parte din laborator

```
clear
close all
clc

s = tf('s');

A = randi(9,4,4)
```

```
A = 4×4
     8     6     9     9
     9     1     9     5
     2     3     2     8
     9     5     9     2
```

```
eig(A)
```

```
ans = 4×1
   23.8750
   -0.5398
   -6.4494
   -3.8857
```

```
B = randi(9,4,3)
```

```
B = 4×3
     4     6     7
     9     1     7
     8     8     7
     9     9     4
```

```
C = randi(9,2,4)
```

```
C = 2×4
     6     7     3     1
     2     1     1     8
```

```
D = randi(9,2,3)
```

```
D = 2×3
     7     9     4
     3     1     4
```

Vad daca este stabilizabil + detectabil

```
Pc = ctrb(A,B);

disp('Rangul matricii de ctrb');
```

```
Rangul matricii de ctrb
```

```
disp(rank(Pc))
```

```
     4
```

```
Po = obsv(A,C);

disp('Rangul matricii de obsv');
```

```
Rangul matricii de obsv
```

```
disp(rank(Po))
```

```
     4
```

Sistem minimal, este bine, pot gasi un K pentru care stabilizez pe G

Youla pe MIMO. Devine interesant

Sa ne readucem aminte la SISO cum arata G-ul: $G = \frac{N}{M}$, unde M si N sunt stabile si coprime intre ele. **Mod de construire:** stim ca $\sigma n \leq \sigma m = k$ si $G = \frac{n}{m}$. In acest sens, $\begin{cases} N = \frac{n}{(s+1)^k} \\ M = \frac{m}{(s+1)^k} \end{cases}$. De asemenea, $\exists X, Y \in \mathcal{S}$ astfel incat $\mathrm{NX} + \mathrm{MY} = 1$, egalitatea lui Bezout. K in forma generica este $K = \left\{ \frac{X + \mathrm{MQ}}{Y + \mathrm{NQ}} \mid Q \in \mathcal{S} \right\}$

La MIMO incep sa se complice lucrurile:

- **Functiile de transfer** (scalare) sunt **inlocuite** acum cu **Matrici de Transfer (MT)**
- **MT nu au** (in general) proprietatea de **comutativitate la inmultire**, iar asta este **critica la scrierea factorizarilor**
- Trebuie **dezvoltata o teorie de factorizari coprime** peste inelul matricelor polinimoale / rationale stabile si **nu exista un Euclid pentru matrice**
- Chiar cu **teoria de divizibilitate** peste matricil polinomiale / rationale stabile disponibila, **o dificultate suplimentara** este data de faptul ca algoritmii rezultati sunt **greu de implementat si numeric instabili**

Macar avem si vesti bune: $\begin{cases} \exists\, o \text{ identitate Bezout peste matrice polinomiale sau rationale in } \mathscr{RH}_\infty \\ \text{Parametrizare Youla se poate scrie din nou in raport cu ea} \end{cases}$

Din cate se poate observa, nu mai avem factotizare coprima $\in \mathscr{S}$. O sa avem nevoie de factorizari la *stanga si dreapta*

Ok, cum definim coprimitatea la MT (rationale)

Fie 2 matrici $M, N \in \mathscr{RH}_\infty$. Acestea sunt coprime daca au acelasi numar de coloane si $\exists \widetilde{X}, \widetilde{Y} \in \mathscr{RH}_\infty$ astfel incat sa aiba loc $\begin{bmatrix} \widetilde{Y} & -\widetilde{X} \end{bmatrix} \begin{bmatrix} M \\ N \end{bmatrix} = \widetilde{Y}M - \widetilde{X}N = I$, iar $G = \mathrm{NM}^{-1}$ (Bezout la dreapta)

Fie alte 2 matrici $\widetilde{M}, \widetilde{N} \in \mathscr{RH}_\infty$. Acestea sunt coprime daca au acelasi numar de linii si $\exists X, Y \in \mathscr{RH}_\infty$ astfel incat ca aiba loc $\begin{bmatrix} \widetilde{N} & \widetilde{M} \end{bmatrix} \begin{bmatrix} -X \\ Y \end{bmatrix} = \widetilde{M}Y - \widetilde{N}X = I$, iar $G = \widetilde{M}^{-1}\widetilde{N}$ (Bezout la stanga)

G are factorilzare **dublu coprima** daca au loc $G = \mathrm{NM}^{-1} = \widetilde{M}^{-1}\widetilde{N}$, iar $N, M, X, Y, \widetilde{N}, \widetilde{M}, \widetilde{X}, \widetilde{Y}$ respecta $\begin{bmatrix} Y & -\widetilde{X} \\ -N & \widetilde{M} \end{bmatrix} \begin{bmatrix} M & X \\ N & Y \end{bmatrix} = I$. Aceasta relatie spune mai mult decat factorizarile coprime la stanga si dreapta si identitatile Bezout corespunzatoare de corpimitate

Fie $\begin{cases} G = \mathrm{NM}^{-1} = \widetilde{M}^{-1}\widetilde{N} \\ K = \mathrm{XY}^{-1} = \widetilde{Y}^{-1}\widetilde{X} \end{cases}$ factorizari coprime la stanga si dreapta pentur sistem si regualtor.

Urmatoarele afirmatii sunt echivalente:

Sistemul ini bucla de reactie este intern stabil:

- $\begin{bmatrix} \mathbf{M} & \mathbf{X} \\ \mathbf{N} & \mathbf{Y} \end{bmatrix}$ *este inversabilă în* $\mathcal{RH}_\infty$;
- $\begin{bmatrix} \widetilde{\mathbf{Y}} & -\widetilde{\mathbf{X}} \\ -\widetilde{\mathbf{N}} & \widetilde{\mathbf{M}} \end{bmatrix}$ *este inversabilă în* $\mathcal{RH}_\infty$;
- $\widetilde{\mathbf{M}}\mathbf{Y} - \widetilde{\mathbf{N}}\mathbf{X}$ *este inversabilă în* $\mathcal{RH}_\infty$;
- $\widetilde{\mathbf{Y}}\mathbf{M} - \widetilde{\mathbf{X}}\mathbf{N}$ *este inversabilă în* $\mathcal{RH}_\infty$.

Sa ne readucem aminte cum arata clasa compenastoarelor la SISO:

$$K = \left\{ \frac{X + \mathrm{MQ}}{Y + \mathrm{NQ}} \,|\, Q \in \mathcal{S} \right\}$$

Acum sa trecem la MIMO:

Pentru $G = \mathrm{NM}^{-1} = \widetilde{M}^{-1}\widetilde{N}$, clasa tuturor compensatoarelor **cu MT proprie** care stabilizeaza intern bucla de reactie este data de una din urmatoarele expresii **echivalente:**

$$\begin{cases} K = (X + \mathrm{MQ})(Y + \mathrm{NQ})^{-1}, \text{unde } Q \in \mathscr{RH}_\infty \text{ cu } \det(Y + \mathrm{NQ})(\infty) \neq 0 \\ K = \left(\widetilde{Y} + Q\widetilde{N}\right)^{-1}\left(\widetilde{X} + Q\widetilde{M}\right), \text{unde } Q \in \mathscr{RH}_\infty \text{ cu } \det\left(\widetilde{Y} + Q\widetilde{N}\right)(\infty) \neq 0 \end{cases}$$, iar

$$\begin{cases} X \text{ si } Y \text{ verifica Bezout } \widetilde{M}Y - \widetilde{X}N = I \\ \widetilde{X} \text{ si } \widetilde{Y} \text{ verifica Bezout } \widetilde{Y}M - \widetilde{X}N = I \end{cases}$$. Daca $X, Y, \widetilde{X}, \widetilde{Y}$ sutn luate astfel incat sa satisfaca factorilzarea dublu coprima $\begin{bmatrix} Y & -\widetilde{X} \\ -N & \widetilde{M} \end{bmatrix}\begin{bmatrix} M & X \\ N & Y \end{bmatrix} = I$ si $\begin{cases} Y \\ \widetilde{Y} \end{cases}$ **sa fie inversabile, de preferat cu inverse proprii**, **atunci K poate fi scris ca TLFi(J,Q), unde** $J = \begin{bmatrix} \mathrm{XY}^{-1} & \widetilde{Y}^{-1} \\ Y^{-1} & -Y^{-1}N \end{bmatrix}$ cu $Q \in \mathscr{RH}_\infty$ astfel incat $\det(Y + \mathrm{NQ})(\infty) \neq 0$.

Daca avem factoriyare dublu coprima pentru G, cat si o factoriyare coprima la dreapta $K = \widehat{X}\widehat{Y}^{-1}$ a unui regulator stabilizator pentru G, atunci parametru $Q \in \mathscr{RH}_\infty$ din Youla este dat de

$Q = M^{-1}\left(\widehat{X}\left(\widehat{M}\widehat{Y} - \widehat{N}\widehat{X}\right)^{-1} - X\right)$. Asta este folositor deoarece se gaseste o solutie cu Youla si matematic sunt sanse sa fi determinat toata clasa

Ok,ok. Cum obtinem o factorizare coprima peste $\mathscr{RH}_\infty$, cum rezolvam identitatea Bezout cu variantele la stanga si la dreapta si cum obtinem o factorizare dublu coprima pentru MIMO? Ne folosim de reprezentarile pe stare (state-space)

Fie $G(s) = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$, cu $\begin{cases} (A, B) \text{ stabilizabil} \\ (C, A) \text{ detectabil} \end{cases}$. In acest sens, fie F si L astfel incat $\begin{cases} \Lambda(A + \mathrm{BF}) \in \mathbb{C}_- \\ \Lambda(A + \mathrm{LC}) \in \mathbb{C}_- \end{cases}$. Atunci

$$\begin{cases} \begin{bmatrix} M & X \\ N & Y \end{bmatrix} = \begin{bmatrix} A + \mathrm{BF} & B & -L \\ F & I & 0 \\ C + \mathrm{DF} & 0 & I \end{bmatrix} \\ \begin{bmatrix} \widetilde{Y} & -\widetilde{X} \\ -\widetilde{N} & \widetilde{M} \end{bmatrix} = \begin{bmatrix} A + \mathrm{LC} & -(B + \mathrm{LD}) & L \\ F & I & 0 \\ C & -D & I \end{bmatrix} \end{cases}$$

$$\begin{bmatrix} \mathbf{M} & \mathbf{X} \\ \mathbf{N} & \mathbf{Y} \end{bmatrix} = \left[\begin{array}{c|cc} A + BF & B & -L \\ \hline F & I & 0 \\ C + DF & D & I \end{array}\right]$$

$$\begin{bmatrix} \widetilde{\mathbf{Y}} & -\widetilde{\mathbf{X}} \\ -\widetilde{\mathbf{N}} & \widetilde{\mathbf{M}} \end{bmatrix} = \left[\begin{array}{c|cc} A + LC & -(B + LD) & L \\ \hline F & I & 0 \\ C & -D & I \end{array}\right]$$

reprezintă o factorizare dublu coprimă (56), (57) a lui **G**.

. Atunci J va avea forma

$$J = \begin{bmatrix} A + \mathrm{BF} + \mathrm{LC} + \mathrm{LDF} & -L & B + \mathrm{LD} \\ F & 0 & I \\ -(C + \mathrm{DF}) & I & -D \end{bmatrix}$$

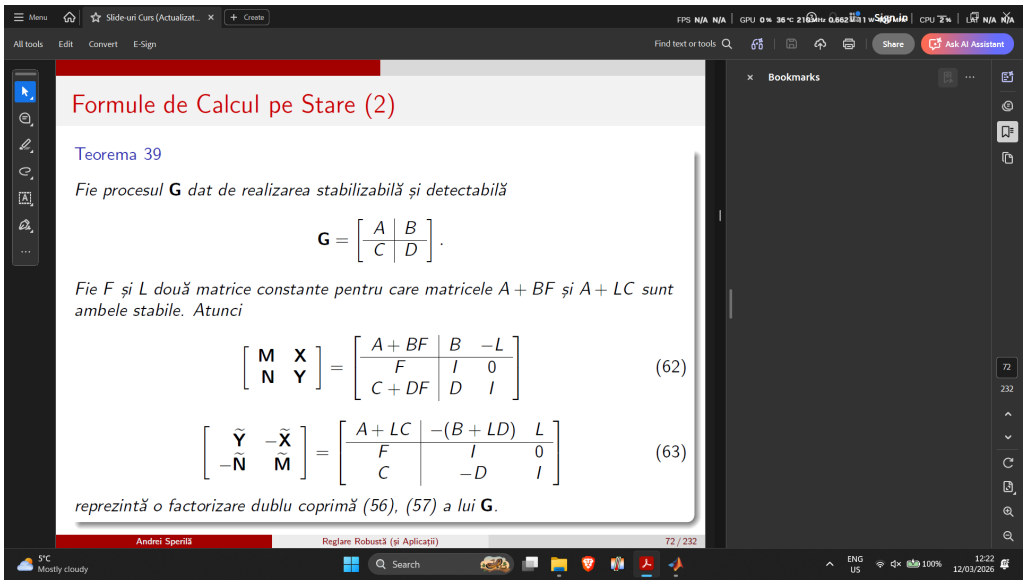
Fie **G** *dat de o realizare de stare ca în Teorema 39, iar (62) și (63) o factorizare dublu coprimă. Atunci* **J** *din (60) se poate calcula ca*

$$\mathbf{J} = \left[\begin{array}{c|cc} A + BF + LC + LDF & -L & B + LD \\ \hline F & 0 & I \\ -(C + DF) & I & -D \end{array}\right].$$

Observam ceva interesant? Daca fac $Q = 0$, atunci $\mathrm{TLFi}(J, Q) = J_{11} + J_{12}Q(I - J_{22}Q)^{-1}J_{21}$. Daca inlocuim pe $Q = 0$ o sa avem $\mathrm{TLFi}(J, 0) = J_{11} = \begin{bmatrix} A + \mathrm{BF} + LC + \mathrm{LDF} & -L \\ F & 0 \end{bmatrix}$ care este **Compensatorul Kalman de la TSA**

Aici intra partea 2

Din exemplul nostru, (A, B) ctrb $\rightarrow$ alocabila, adica pot aloca polii unde vreau eu folosind un F, adica $\Lambda(A + \mathrm{BF}) \in \mathbb{C}_-$. Pentru F real, spectru simetric



Unde F si L asigura $\Lambda(A + \mathrm{BF})$ si $\Lambda(A + \mathrm{LC}) \in \mathbb{C}_-$

```
F = -place(A,B,[-1 -1 -1 -2])
```

```
F = 3×4
   -0.5000    0.1681   -0.4790    0.5503
    0.0319   -0.4409   -0.0071   -0.2871
   -0.5137   -0.3908   -0.5501   -1.3901
```

```
L = -place(A',C',[-1 -1 -2 -2])'
```

```
L = 4×2
   27.7497  -39.7132
   37.3043  -51.8154
    5.7688   -9.0203
   31.7225  -44.4243
```

```
disp('Valori proprii ale lui A+BF');
```

```
Valori proprii ale lui A+BF
```

```
eig(A + B*F)
```

```
ans = 4×1
   -2.0000
   -1.0000
   -1.0000
   -1.0000
```

```
disp('Valori proprii ale lui A+LC');
```

```
Valori proprii ale lui A+LC
```

```
eig(A + L*C)
```

```
ans = 4×1
   -1.0000
   -2.0000
   -2.0000
   -1.0000
```

```
MatStanga = [A + B*F B -L;
             F eye(3) zeros(3,2)
             C -D eye(2,2)]
```

```
MatStanga = 9×9
    2.5952    1.2913    3.1905   -0.2522    4.0000    6.0000    7.0000  -27.7497 ···
    0.9360   -0.6638    0.8306   -0.0657    9.0000    1.0000    7.0000  -37.3043
   -5.3410   -1.9183   -5.7397    0.3747    8.0000    8.0000    7.0000   -5.7688
    2.7321    0.9813    2.4245   -1.1917    9.0000    9.0000    4.0000  -31.7225
   -0.5000    0.1681   -0.4790    0.5503    1.0000         0         0         0
    0.0319   -0.4409   -0.0071   -0.2871         0    1.0000         0         0
   -0.5137   -0.3908   -0.5501   -1.3901         0         0    1.0000         0
    6.0000    7.0000    3.0000    1.0000   -7.0000   -9.0000   -4.0000    1.0000
    2.0000    1.0000    1.0000    8.0000   -3.0000   -1.0000   -4.0000         0
```

```
MatDreapta = [A + L*C -(B + L*D) L
                F eye(3) zeros(3,2)
                C -D eye(2)]
```

```
MatDreapta = 9×9
   95.0718  160.5347   52.5359 -280.9558  -79.1083 -216.0341   40.8540   27.7497 ···
  129.1947  210.3144   69.0974 -372.2191 -114.6836 -284.9229   51.0446   37.3043
   18.5720   34.3611   10.2860  -58.3935  -21.3205  -50.8986    6.0061    5.7688
  110.4864  182.6333   59.7432 -321.6722  -97.7846 -250.0783   46.8073   31.7225
   -0.5000    0.1681   -0.4790    0.5503    1.0000         0         0         0
    0.0319   -0.4409   -0.0071   -0.2871         0    1.0000         0         0
   -0.5137   -0.3908   -0.5501   -1.3901         0         0    1.0000         0
    6.0000    7.0000    3.0000    1.0000   -7.0000   -9.0000   -4.0000    1.0000
    2.0000    1.0000    1.0000    8.0000   -3.0000   -1.0000   -4.0000         0
```

Acum definesc cele 8 matrici bloc

Tilda

```
Ytilde = ss(A + L*C, -(B + L*D), F, eye(3))
```

```
Ytilde =

  A =
           x1      x2      x3      x4
   x1   95.07   160.5   52.54    -281
   x2   129.2   210.3    69.1  -372.2
   x3   18.57   34.36   10.29  -58.39
   x4   110.5   182.6   59.74  -321.7

  B =
           u1      u2      u3
   x1  -79.11    -216   40.85
   x2  -114.7  -284.9   51.04
   x3  -21.32   -50.9   6.006
```

```
  x4  -97.78  -250.1   46.81

  C =
             x1        x2        x3        x4
  y1       -0.5    0.1681    -0.479    0.5503
  y2    0.03187   -0.4409  -0.00706   -0.2871
  y3    -0.5137   -0.3908   -0.5501     -1.39

  D =
       u1  u2  u3
  y1    1   0   0
  y2    0   1   0
  y3    0   0   1

Continuous-time state-space model.
Model Properties
```

```
Xtilde = -ss(A + L*C, L, F, zeros(3,2))
```

```
Xtilde =

  A =
           x1      x2      x3      x4
  x1    95.07   160.5   52.54    -281
  x2    129.2   210.3    69.1  -372.2
  x3    18.57   34.36   10.29  -58.39
  x4    110.5   182.6   59.74  -321.7

  B =
           u1      u2
  x1    27.75  -39.71
  x2     37.3  -51.82
  x3    5.769   -9.02
  x4    31.72  -44.42

  C =
             x1        x2        x3        x4
  y1        0.5   -0.1681     0.479   -0.5503
  y2   -0.03187    0.4409   0.00706    0.2871
  y3     0.5137    0.3908    0.5501      1.39

  D =
       u1  u2
  y1    0   0
  y2    0   0
  y3    0   0

Continuous-time state-space model.
Model Properties
```

```
Ntilde = -ss(A + L*C, -(B + L*D),C, -D)
```

```
Ntilde =

  A =
           x1      x2      x3      x4
  x1    95.07   160.5   52.54    -281
  x2    129.2   210.3    69.1  -372.2
  x3    18.57   34.36   10.29  -58.39
  x4    110.5   182.6   59.74  -321.7
```

```
  B =
           u1      u2      u3
   x1  -79.11    -216   40.85
   x2  -114.7  -284.9   51.04
   x3  -21.32   -50.9   6.006
   x4  -97.78  -250.1   46.81

  C =
       x1  x2  x3  x4
   y1  -6  -7  -3  -1
   y2  -2  -1  -1  -8

  D =
       u1  u2  u3
   y1   7   9   4
   y2   3   1   4

Continuous-time state-space model.
Model Properties
```

```
Mtilde = ss(A + L*C, L, C, eye(2))
```

```
Mtilde =

  A =
           x1      x2      x3      x4
   x1   95.07   160.5   52.54    -281
   x2   129.2   210.3    69.1  -372.2
   x3   18.57   34.36   10.29  -58.39
   x4   110.5   182.6   59.74  -321.7

  B =
           u1      u2
   x1   27.75  -39.71
   x2    37.3  -51.82
   x3   5.769   -9.02
   x4   31.72  -44.42

  C =
       x1  x2  x3  x4
   y1   6   7   3   1
   y2   2   1   1   8

  D =
       u1  u2
   y1   1   0
   y2   0   1

Continuous-time state-space model.
Model Properties
```

```
ML = [Ytilde, -Xtilde;
      -Ntilde, Mtilde]
```

```
ML =

  A =
            x1      x2      x3      x4      x5      x6      x7      x8      x9     x10     x11     x12     x13     x
   x1    95.07   160.5   52.54    -281       0       0       0       0       0       0       0       0       0
   x2    129.2   210.3    69.1  -372.2       0       0       0       0       0       0       0       0       0
```

```
  x3    18.57   34.36   10.29  -58.39       0       0       0       0       0       0       0       0       0
  x4    110.5   182.6   59.74  -321.7       0       0       0       0       0       0       0       0       0
  x5        0       0       0       0   95.07   160.5   52.54    -281       0       0       0       0       0
  x6        0       0       0       0   129.2   210.3    69.1  -372.2       0       0       0       0       0
  x7        0       0       0       0   18.57   34.36   10.29  -58.39       0       0       0       0       0
  x8        0       0       0       0   110.5   182.6   59.74  -321.7       0       0       0       0       0
  x9        0       0       0       0       0       0       0       0   95.07   160.5   52.54    -281       0
  x10       0       0       0       0       0       0       0       0   129.2   210.3    69.1  -372.2       0
  x11       0       0       0       0       0       0       0       0   18.57   34.36   10.29  -58.39       0
  x12       0       0       0       0       0       0       0       0   110.5   182.6   59.74  -321.7       0
  x13       0       0       0       0       0       0       0       0       0       0       0       0   95.07   166
  x14       0       0       0       0       0       0       0       0       0       0       0       0   129.2   216
  x15       0       0       0       0       0       0       0       0       0       0       0       0   18.57   34.
  x16       0       0       0       0       0       0       0       0       0       0       0       0   110.5   182

B =
          u1      u2      u3      u4      u5
  x1   -79.11    -216   40.85       0       0
  x2   -114.7  -284.9   51.04       0       0
  x3   -21.32   -50.9   6.006       0       0
  x4   -97.78  -250.1   46.81       0       0
  x5        0       0       0   27.75  -39.71
  x6        0       0       0    37.3  -51.82
  x7        0       0       0   5.769   -9.02
  x8        0       0       0   31.72  -44.42
  x9   -79.11    -216   40.85       0       0
  x10  -114.7  -284.9   51.04       0       0
  x11  -21.32   -50.9   6.006       0       0
  x12  -97.78  -250.1   46.81       0       0
  x13       0       0       0   27.75  -39.71
  x14       0       0       0    37.3  -51.82
  x15       0       0       0   5.769   -9.02
  x16       0       0       0   31.72  -44.42

C =
           x1        x2        x3        x4        x5        x6        x7        x8        x9       x10       x11
  y1      -0.5    0.1681    -0.479    0.5503      -0.5    0.1681    -0.479    0.5503         0         0         0
  y2   0.03187   -0.4409  -0.00706   -0.2871   0.03187   -0.4409  -0.00706   -0.2871         0         0         0
  y3   -0.5137   -0.3908   -0.5501     -1.39   -0.5137   -0.3908   -0.5501     -1.39         0         0         0
  y4         0         0         0         0         0         0         0         0         6         7         3
  y5         0         0         0         0         0         0         0         0         2         1         1

D =
      u1  u2  u3  u4  u5
  y1   1   0   0   0   0
  y2   0   1   0   0   0
  y3   0   0   1   0   0
  y4  -7  -9  -4   1   0
  y5  -3  -1  -4   0   1

Continuous-time state-space model.
Model Properties
```

```
size(ML)
```

```
State-space model with 5 outputs, 5 inputs, and 16 states.
```

NeTilda

```
M = ss(A + B*F, B, F, eye(3))
```

```
M =
```

```
  A =
             x1        x2        x3        x4
   x1     2.595     1.291     3.191   -0.2522
   x2     0.936   -0.6638    0.8306  -0.06567
   x3    -5.341    -1.918     -5.74    0.3747
   x4     2.732    0.9813     2.425    -1.192

  B =
       u1  u2  u3
   x1   4   6   7
   x2   9   1   7
   x3   8   8   7
   x4   9   9   4

  C =
             x1        x2        x3        x4
   y1      -0.5    0.1681    -0.479    0.5503
   y2   0.03187   -0.4409  -0.00706   -0.2871
   y3   -0.5137   -0.3908   -0.5501     -1.39

  D =
       u1  u2  u3
   y1   1   0   0
   y2   0   1   0
   y3   0   0   1

Continuous-time state-space model.
Model Properties
```

```
N = ss(A + B*F, B, C + D*F, D)
```

```
N =

  A =
             x1        x2        x3        x4
   x1     2.595     1.291     3.191   -0.2522
   x2     0.936   -0.6638    0.8306  -0.06567
   x3    -5.341    -1.918     -5.74    0.3747
   x4     2.732    0.9813     2.425    -1.192

  B =
       u1  u2  u3
   x1   4   6   7
   x2   9   1   7
   x3   8   8   7
   x4   9   9   4

  C =
            x1       x2       x3       x4
   y1   0.7321    2.645   -2.617   -3.292
   y2   -1.523  -0.4999   -2.645    3.803

  D =
       u1  u2  u3
   y1   7   9   4
   y2   3   1   4

Continuous-time state-space model.
Model Properties
```

```
X = ss(A + B*F, -L, F, zeros(3,2))
```

```
X =

  A =
              x1        x2        x3        x4
   x1     2.595     1.291     3.191   -0.2522
   x2     0.936   -0.6638    0.8306  -0.06567
   x3    -5.341    -1.918     -5.74    0.3747
   x4     2.732    0.9813     2.425    -1.192

  B =
            u1      u2
   x1  -27.75   39.71
   x2   -37.3   51.82
   x3  -5.769    9.02
   x4  -31.72   44.42

  C =
              x1        x2        x3        x4
   y1      -0.5    0.1681    -0.479    0.5503
   y2   0.03187   -0.4409  -0.00706   -0.2871
   y3   -0.5137   -0.3908   -0.5501     -1.39

  D =
       u1  u2
   y1   0   0
   y2   0   0
   y3   0   0

Continuous-time state-space model.
Model Properties
```

```
Y = ss(A + B*F, -L, C + D*F, eye(2))
```

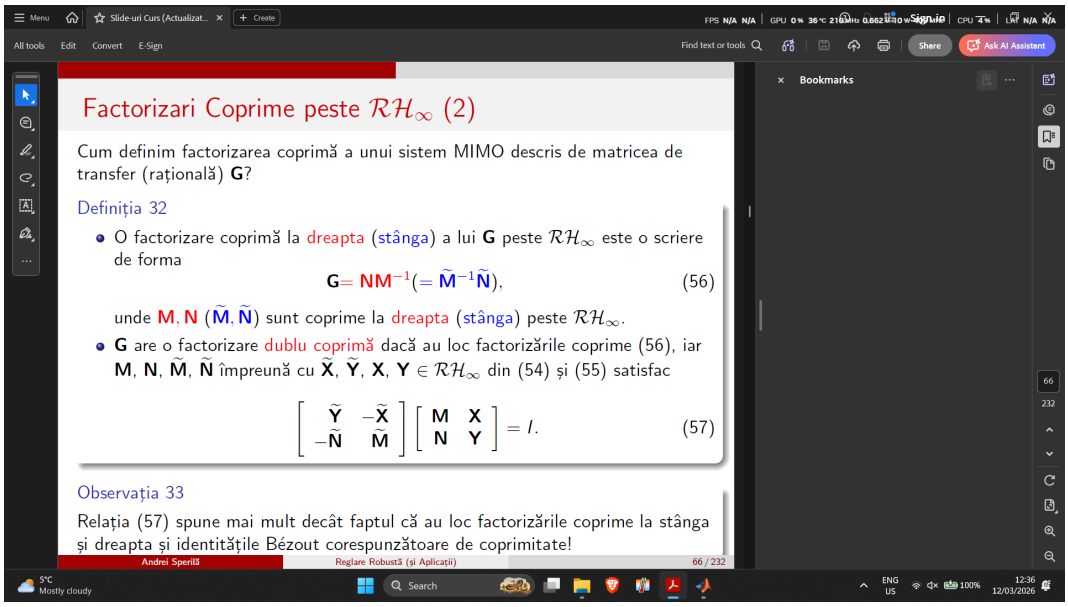
```
Y =

  A =
              x1        x2        x3        x4
   x1     2.595     1.291     3.191   -0.2522
   x2     0.936   -0.6638    0.8306  -0.06567
   x3    -5.341    -1.918     -5.74    0.3747
   x4     2.732    0.9813     2.425    -1.192

  B =
            u1      u2
   x1  -27.75   39.71
   x2   -37.3   51.82
   x3  -5.769    9.02
   x4  -31.72   44.42

  C =
             x1       x2       x3       x4
   y1   0.7321    2.645   -2.617   -3.292
   y2   -1.523  -0.4999   -2.645    3.803

  D =
       u1  u2
   y1   1   0
   y2   0   1

Continuous-time state-space model.
Model Properties
```

```
MR = [M,X;
      N,Y]
```

```
MR =

  A =
              x1        x2        x3        x4        x5        x6        x7        x8        x9       x10       x11
   x1      2.595     1.291     3.191   -0.2522         0         0         0         0         0         0         0
   x2      0.936   -0.6638    0.8306  -0.06567         0         0         0         0         0         0         0
   x3     -5.341    -1.918     -5.74    0.3747         0         0         0         0         0         0         0
   x4      2.732    0.9813     2.425    -1.192         0         0         0         0         0         0         0
   x5          0         0         0         0     2.595     1.291     3.191   -0.2522         0         0         0
   x6          0         0         0         0     0.936   -0.6638    0.8306  -0.06567         0         0         0
   x7          0         0         0         0    -5.341    -1.918     -5.74    0.3747         0         0         0
   x8          0         0         0         0     2.732    0.9813     2.425    -1.192         0         0         0
   x9          0         0         0         0         0         0         0         0     2.595     1.291     3.191
   x10         0         0         0         0         0         0         0         0     0.936   -0.6638    0.8306
   x11         0         0         0         0         0         0         0         0    -5.341    -1.918     -5.74
   x12         0         0         0         0         0         0         0         0     2.732    0.9813     2.425
   x13         0         0         0         0         0         0         0         0         0         0         0
   x14         0         0         0         0         0         0         0         0         0         0         0
   x15         0         0         0         0         0         0         0         0         0         0         0
   x16         0         0         0         0         0         0         0         0         0         0         0

  B =
             u1      u2      u3      u4      u5
   x1         4       6       7       0       0
   x2         9       1       7       0       0
   x3         8       8       7       0       0
   x4         9       9       4       0       0
   x5         0       0       0  -27.75   39.71
   x6         0       0       0   -37.3   51.82
   x7         0       0       0  -5.769    9.02
   x8         0       0       0  -31.72   44.42
   x9         4       6       7       0       0
   x10        9       1       7       0       0
   x11        8       8       7       0       0
   x12        9       9       4       0       0
   x13        0       0       0  -27.75   39.71
   x14        0       0       0   -37.3   51.82
   x15        0       0       0  -5.769    9.02
   x16        0       0       0  -31.72   44.42

  C =
              x1        x2        x3        x4        x5        x6        x7        x8        x9       x10       x11
   y1       -0.5    0.1681    -0.479    0.5503      -0.5    0.1681    -0.479    0.5503         0         0         0
   y2    0.03187   -0.4409  -0.00706   -0.2871   0.03187   -0.4409  -0.00706   -0.2871         0         0         0
   y3    -0.5137   -0.3908   -0.5501     -1.39   -0.5137   -0.3908   -0.5501     -1.39         0         0         0
   y4          0         0         0         0         0         0         0         0    0.7321     2.645    -2.617
   y5          0         0         0         0         0         0         0         0    -1.523   -0.4999    -2.645

  D =
        u1  u2  u3  u4  u5
   y1    1   0   0   0   0
   y2    0   1   0   0   0
   y3    0   0   1   0   0
   y4    7   9   4   1   0
   y5    3   1   4   0   1

Continuous-time state-space model.
Model Properties
```

```
size(MR)
```

```
State-space model with 5 outputs, 5 inputs, and 16 states.
```



Ma pot uita la $G * M - N$ si $\widetilde{M} * G - \widetilde{N}$ pentru a vedea daca chiar este coprima factorizarea

$G = \mathrm{NM}^{-1} = \widetilde{M}^{-1}\widetilde{N}$

```
G = ss(A,B,C,D)
```

```
G =

  A =
       x1  x2  x3  x4
   x1   8   6   9   9
   x2   9   1   9   5
   x3   2   3   2   8
   x4   9   5   9   2

  B =
       u1  u2  u3
   x1   4   6   7
   x2   9   1   7
   x3   8   8   7
   x4   9   9   4

  C =
       x1  x2  x3  x4
   y1   6   7   3   1
   y2   2   1   1   8

  D =
       u1  u2  u3
   y1   7   9   4
   y2   3   1   4

Continuous-time state-space model.
Model Properties
```

```
norm(G * M - N, 'inf')
```

```
ans =
2.3795e-13
```

```
norm(Mtilde * G - Ntilde, 'inf')
```

```
ans =
8.6446e-10
```

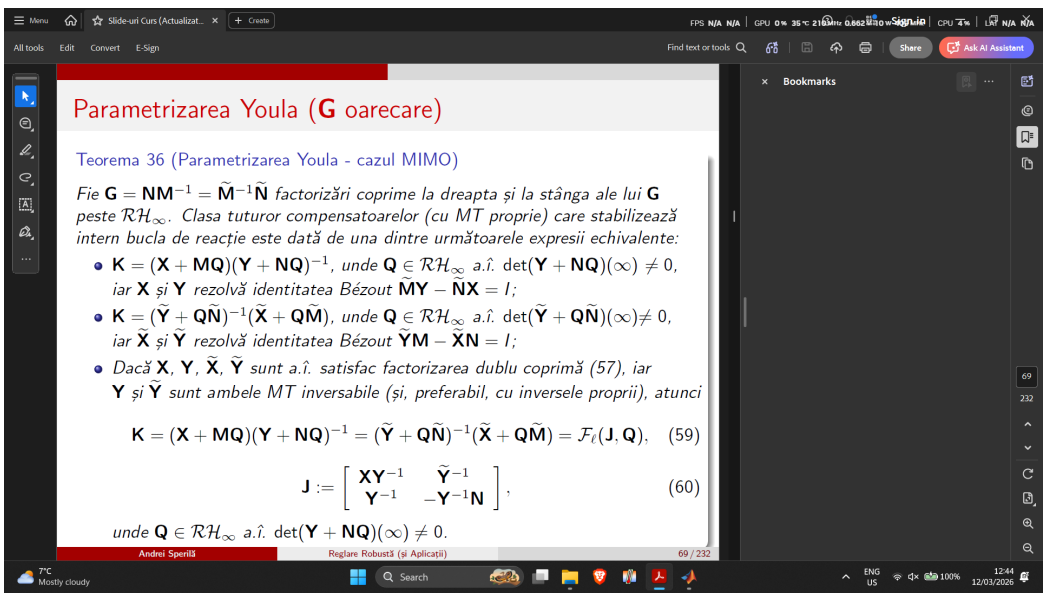
Acum verific Bezout

```
norm(ML * MR - eye(5), 'inf')
```

```
ans =
3.7157e-08
```

```
norm(MR * ML - eye(5), 'inf')
```

```
ans =
3.8338e-10
```



Practic este TLFi intre J si K, J sus si K jos

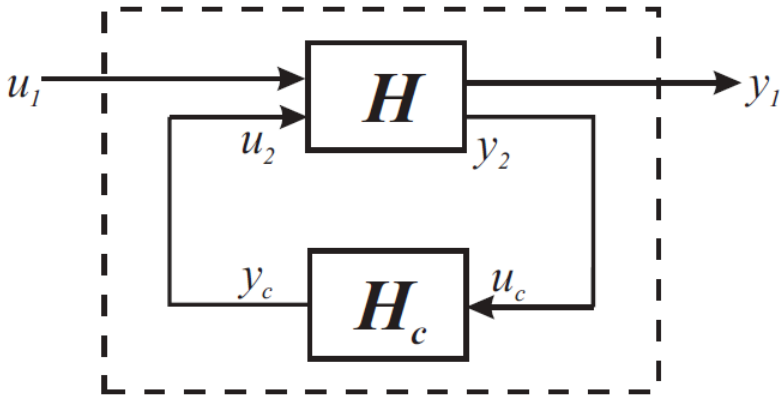


Figura 5: Transformare liniar fracționară inferioară

```
J = ss(A + B*F + L*C + L*D*F, [-L, B + L*D], ...
    [F; -(C + D*F)], [zeros(3,2), eye(3,3);eye(2,2), -D])
```

```
J =

  A =
           x1      x2      x3      x4
   x1   83.39   94.54   35.59  -242.6
   x2   107.2   123.9   40.23  -319.9
   x3   12.62   17.85   3.017  -52.92
   x4   93.61   107.1   36.88  -274.6

  B =
           u1      u2      u3      u4      u5
   x1  -27.75   39.71   79.11     216  -40.85
   x2   -37.3   51.82   114.7   284.9  -51.04
   x3  -5.769    9.02   21.32    50.9  -6.006
   x4  -31.72   44.42   97.78   250.1  -46.81

  C =
             x1        x2        x3        x4
   y1      -0.5    0.1681    -0.479    0.5503
   y2   0.03187   -0.4409  -0.00706   -0.2871
   y3   -0.5137   -0.3908   -0.5501     -1.39
   y4   -0.7321    -2.645     2.617     3.292
   y5     1.523    0.4999     2.645    -3.803

  D =
       u1  u2  u3  u4  u5
   y1   0   0   1   0   0
   y2   0   0   0   1   0
   y3   0   0   0   0   1
   y4   1   0  -7  -9  -4
   y5   0   1  -3  -1  -4

Continuous-time state-space model.
Model Properties
```

```
Q = zeros(3,2)
```

```
Q = 3×2
     0     0
```

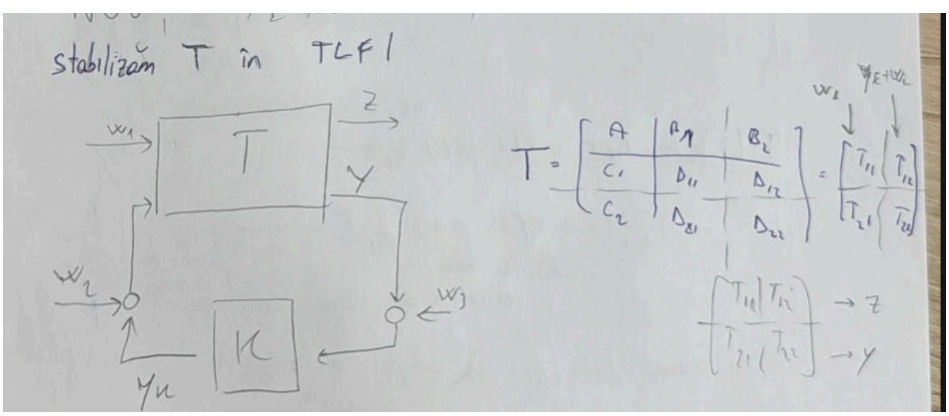
```
     0     0
     0     0
```

```
Kc = lft(J,Q) % strict propriu, conexiune bine definita
```

```
Kc =

  A =
           x1      x2      x3      x4
   x1   83.39   94.54   35.59  -242.6
   x2   107.2   123.9   40.23  -319.9
   x3   12.62   17.85   3.017  -52.92
   x4   93.61   107.1   36.88  -274.6

  B =
           u1      u2
   x1  -27.75   39.71
   x2   -37.3   51.82
   x3  -5.769    9.02
   x4  -31.72   44.42

  C =
             x1        x2        x3        x4
   y1      -0.5    0.1681    -0.479    0.5503
   y2   0.03187   -0.4409  -0.00706   -0.2871
   y3   -0.5137   -0.3908   -0.5501     -1.39

  D =
       u1  u2
   y1   0   0
   y2   0   0
   y3   0   0

Continuous-time state-space model.
Model Properties
```

```
eig(Kc) % depinde cum nimeresti, de obicei nu este stabil
```

```
ans = 4×1
  -46.8547
  -15.8003
   -1.0000
   -0.6077
```



Remarca: Ce este cu linie punctata sunt pentru state-space

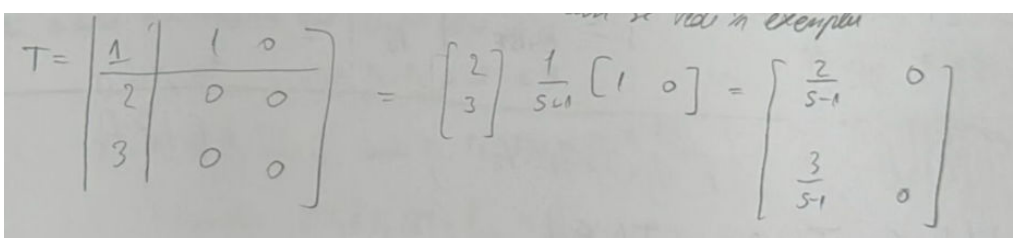
Egalitatea $\begin{bmatrix} A & B_1 & B_2 \\ C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & D_{22} \end{bmatrix} = \begin{bmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{bmatrix}$ are loc $\Leftrightarrow D_{22}$ conectat la K

Fie $K = \begin{bmatrix} A_K & B_K \\ C_K & D_K \end{bmatrix}$. Daca $\begin{cases} (A, B_2) \text{ stabilizabil} \\ (C_2, A) \text{ detectabil} \\ (A_K, B_K) \text{ stabilizabil} \\ (C_K, A_K) \text{ detectabil} \end{cases} \Rightarrow$ stabilizarea pe stare in TLFi **este echivalenta** cu stabilizarea I/O in TFLi sau intre T_{22} si K. Se poate face astfel incat toti polii instabili ai lui T sa fie in T_{22}. Asta ne spune asa: **Daca** T_{22} este stabilizabil si detectabil, atunci si T este stabilizabil si detectabil. <u>*RECIPROCA NU ESTE ADEVARATA*</u>

In plus, $\begin{bmatrix} I & -D_{22} \\ -D_K & I \end{bmatrix}$ inversabila

In acest sens, avem $\begin{bmatrix} z \\ y \\ y_K \end{bmatrix} = T_{\text{cl}} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix}$ cu $T_{\text{cl}_{11}} = \text{TLFi}(T, K) = T_{11} + T_{12}K(I - T_{22}K)^{-1}T_{21}$

Sa luam niste exemple: $T = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 0 & 0 \\ 3 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \frac{1}{s-1} [1 \quad 0] = \begin{bmatrix} \frac{2}{s-1} & 0 \\ \frac{3}{s-1} & 0 \end{bmatrix}$

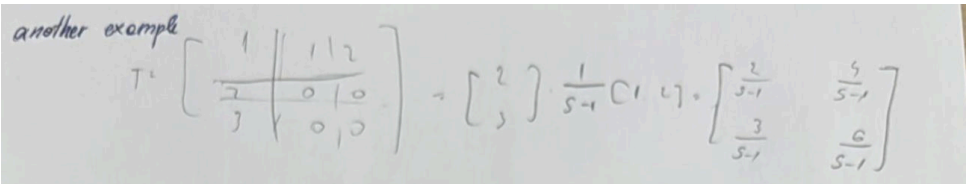


Aici se vede ca T_{22} nu poate vede nimic. Pentru a ne convinge, facem

$\text{TLFi}(T, K) = T_{11} + T_{12}K(I - T_{22}K)^{-1}T_{21} = \frac{2}{s-1} + 0 * K(I - 0 * K)^{-1}\frac{3}{s-1} = \frac{2}{s-1}$ care este mereu instabil, orice K as lua eu (face allahu ahbar boom)

Un ale exemplu

$$T = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 0 & 0 \\ 3 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \frac{1}{s-1} [1 \quad 2] = \begin{bmatrix} \frac{2}{s-1} & \frac{4}{s-1} \\ \frac{3}{s-1} & \frac{6}{s-1} \end{bmatrix}$$

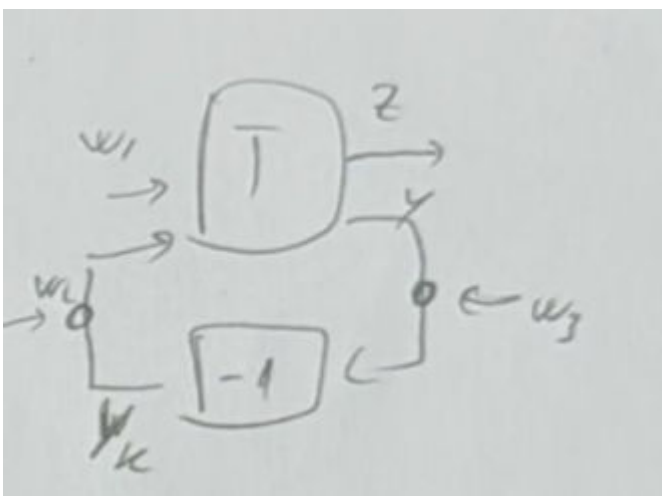


$T_{22} = \frac{6}{s-1}$. Deci putem doar sa stabilizan pe el. Fie un q astfel incat $1 - \frac{6 * q}{s-1} = \frac{s-1-6q}{s-1}$ sa aiba zero stabil.

Pentru $q = -1$ avem $\frac{s+5}{s-1}$ care are zero stabil,adica polul in bucla inchisa

Cum reprezentam asta pe stare? Uite asa:

$$\begin{cases} \Sigma_T \begin{cases} \dot{x} = x + u_1 + 2u_2 \\ y_1 = 2x \\ y_2 = 3x \end{cases} \\ \Sigma_{cl} \begin{cases} \dot{x} = x + w_1 + 2(w_2 + y_k) \\ z = 2x \\ y_k = -w_3 - 3x \end{cases} \end{cases}, \text{ unde } \begin{cases} u_1 = w_1 \\ u_2 = w_2 + y_k \\ y_1 = z \end{cases}$$



Cum stabilizeaza:

```
T_cl = feedback(G,-Kc) % Fortam reactie pozitiva
```

```
T_cl =

  A =
```

```
          x1      x2      x3      x4      x5      x6      x7      x8
   x1      8       6       9       9  -5.405  -4.709  -5.809  -9.252
   x2      9       1       9       5  -8.064  -1.664  -8.169  -5.066
   x3      2       3       2       8  -7.341  -4.918   -7.74  -7.625
   x4      9       5       9       2  -6.268  -4.019  -6.575  -3.192
   x5 -87.07  -154.5  -43.54     290   89.67   155.8   46.73  -290.2
   x6 -120.2  -209.3   -60.1   377.2   121.1   208.7   60.93  -377.3
   x7 -16.57  -31.36  -8.286   66.39   11.23   29.44   2.546  -66.02
   x8 -101.5  -177.6  -50.74   323.7   104.2   178.6   53.17  -324.9

  B =
          u1      u2      u3
   x1      4       6       7
   x2      9       1       7
   x3      8       8       7
   x4      9       9       4
   x5 -75.11    -210   47.85
   x6 -105.7  -283.9   58.04
   x7 -13.32   -42.9   13.01
   x8 -88.78  -241.1   50.81

  C =
          x1      x2      x3      x4      x5      x6      x7      x8
   y1      6       7       3       1  -5.268  -4.355  -5.617  -4.292
   y2      2       1       1       8  -3.523    -1.5  -3.645  -4.197

  D =
       u1  u2  u3
   y1   7   9   4
   y2   3   1   4

Continuous-time state-space model.
Model Properties
```

```
disp('Valorile proprii ale buclei inchise')
```

```
Valorile proprii ale buclei inchise
```

```
eig(T_cl) % sa traiasca Principiul Separatiei
```

```
ans = 8×1
   -1.9999
   -2.0001
   -2.0000
   -1.0001
   -0.9999
   -1.0000
   -1.0000
   -1.0000
```

```
disp('Valori proprii ale luI A+BF')
```

```
Valori proprii ale luI A+BF
```

```
eig(A+B*F)
```

```
ans = 4×1
   -2.0000
   -1.0000
   -1.0000
   -1.0000
```

```
disp('Valori proprii ale lui A+LC')
```

```
Valori proprii ale lui A+LC
```

```
eig(A+L*C)
```

```
ans = 4×1
   -1.0000
   -2.0000
   -2.0000
   -1.0000
```

Se poate observa ca $\Lambda(A+\mathrm{BF})$ si $\Lambda(A+\mathrm{LC})$ sunt in $\mathrm{eig}(T_{\mathrm{cl}})$

Aici se schimba Q ala

```
Q = tf(1,[1 3]) * randi(9,3,2)
```

```
Q =

  From input 1 to output...
         7
   1:  -----
       s + 3

         8
   2:  -----
       s + 3

         2
   3:  -----
       s + 3

  From input 2 to output...
         5
   1:  -----
       s + 3

         5
   2:  -----
       s + 3

         6
   3:  -----
       s + 3

Continuous-time transfer function.
Model Properties
```

```
disp('Valorile proprii ale lui Q');
```

```
Valorile proprii ale lui Q
```

```
eig(Q)
```

```
ans = 2×1
    -3
    -3
```

```
K = lft(J,Q)
```

```
K =

  A =
           x1      x2      x3      x4      x5      x6
   x1   83.39   94.54   35.59  -242.6   550.1   307.6
   x2   107.2   123.9   40.23  -319.9     745   422.9
   x3   12.62   17.85   3.017  -52.92   136.1   81.26
   x4   93.61   107.1   36.88  -274.6   647.9   364.6
   x5  -2.928  -10.58   10.47   13.17    -132    -104
   x6   6.092   1.999   10.58  -15.21     -37     -47

  B =
           u1      u2
   x1  -27.75   39.71
   x2   -37.3   51.82
   x3  -5.769    9.02
   x4  -31.72   44.42
   x5       4       0
   x6       0       4

  C =
              x1        x2        x3        x4        x5        x6
   y1       -0.5    0.1681    -0.479    0.5503      1.75      1.25
   y2    0.03187   -0.4409  -0.00706   -0.2871         2      1.25
   y3    -0.5137   -0.3908   -0.5501     -1.39       0.5       1.5

  D =
       u1  u2
   y1   0   0
   y2   0   0
   y3   0   0

Continuous-time state-space model.
Model Properties
```

```
eig(K)
```

```
ans = 6×1
 -137.9577
  -86.2609
  -16.3562
    2.6185
   -1.0000
   -4.3064
```

```
svd(eye(2) - D * K.d)
```

```
ans = 2×1
     1
     1
```

```
T_cl = feedback(G, -K) % Asa fortam reactie pozitiva
```

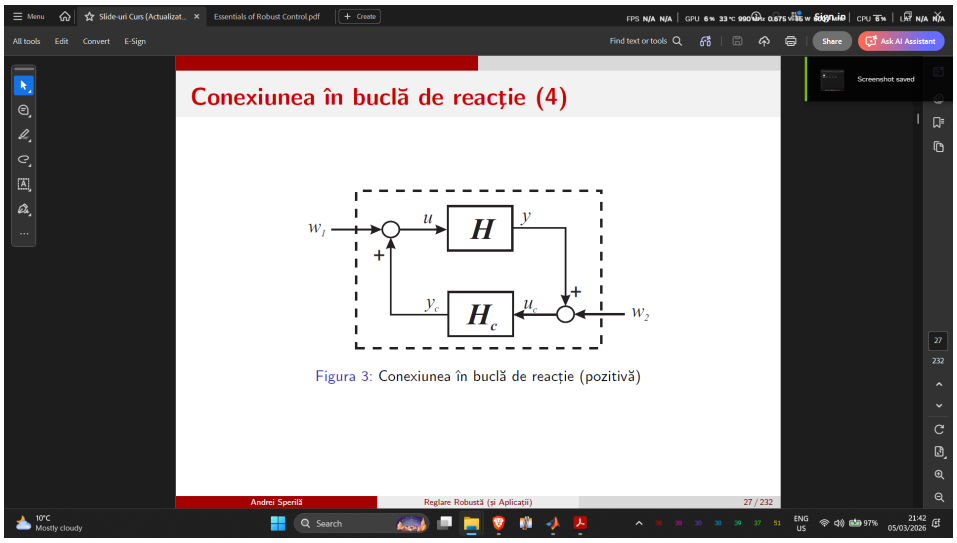
```
T_cl =
```

```
  A =
            x1      x2      x3      x4      x5      x6      x7      x8      x9     x10
   x1        8       6       9       9  -5.405  -4.709  -5.809  -9.252    22.5      23
   x2        9       1       9       5  -8.064  -1.664  -8.169  -5.066   21.25      23
   x3        2       3       2       8  -7.341  -4.918   -7.74  -7.625    33.5    30.5
   x4        9       5       9       2  -6.268  -4.019  -6.575  -3.192   35.75    28.5
   x5   -87.07  -154.5  -43.54     290   89.67   155.8   46.73  -290.2    22.5      23
   x6   -120.2  -209.3   -60.1   377.2   121.1   208.7   60.93  -377.3   21.25      23
   x7   -16.57  -31.36  -8.286   66.39   11.23   29.44   2.546  -66.02    33.5    30.5
   x8   -101.5  -177.6  -50.74   323.7   104.2   178.6   53.17  -324.9   35.75    28.5
   x9       24      28      12       4     -24     -28     -12      -4      -3       0
   x10       8       4       4      32      -8      -4      -4     -32       0      -3

  B =
            u1      u2      u3
   x1        4       6       7
   x2        9       1       7
   x3        8       8       7
   x4        9       9       4
   x5   -75.11    -210   47.85
   x6   -105.7  -283.9   58.04
   x7   -13.32   -42.9   13.01
   x8   -88.78  -241.1   50.81
   x9       28      36      16
   x10      12       4      16

  C =
           x1      x2      x3      x4      x5      x6      x7      x8      x9     x10
   y1       6       7       3       1  -5.268  -4.355  -5.617  -4.292   32.25      26
   y2       2       1       1       8  -3.523    -1.5  -3.645  -4.197    9.25      11

  D =
       u1  u2  u3
   y1   7   9   4
   y2   3   1   4

Continuous-time state-space model.
Model Properties
```

```
eig(T_cl)
```

```
ans = 10×1
   -2.0003
   -1.9997
   -3.0000
   -3.0000
   -2.0000
   -1.0002
   -0.9998
   -1.0000
   -1.0000
   -1.0000
```

```
% observam ca se adauga modurile stabile din Q in T_cl
% fata de inainte
% principiul separatiei sa traiasca again
```




```
clear
close all
clc
```

Ok ok, posibil sa fi vazut ca am mentionat **Principiul Separatiei.** Ce naiba este ala va intrebati

<u>Noi trebuie sa avem un sistem stabilizabil si detectabil pentru asta sa mearga</u>

<u>Intr-un fel separam problema nostra principala in 2: determinarea unui observator (L ala) si determinarea lui F ala</u>

O sa explic in detaliu de unde aceasta "imparteala" (vom considera $D = 0$):

Ce scop avem noi: $u = -F\hat{x}$ sa mentinem stabilitatea sistemului in bucvla inchisa si a observatorului

Mai intai sa luam sistemul nostru original liniar pe stare: $\begin{cases} \dot{x} = \mathrm{Ax} + \mathrm{Bu} \\ y = \mathrm{Cx} \end{cases}$

O sa mai luam si dinamica data de observator: $\dot{\hat{x}} = A\hat{x} + B\hat{u} + L(y - C\hat{x})$. Stim ca

$u = -F\hat{x} \Rightarrow \dot{\hat{x}} = A\hat{x} - \mathrm{BF}\hat{x} + \mathrm{Ly} - \mathrm{LC}\hat{x} = (A - \mathrm{BF} - \mathrm{LC})\hat{x} + \mathrm{Ly}$ cu $u = -F\hat{x}$

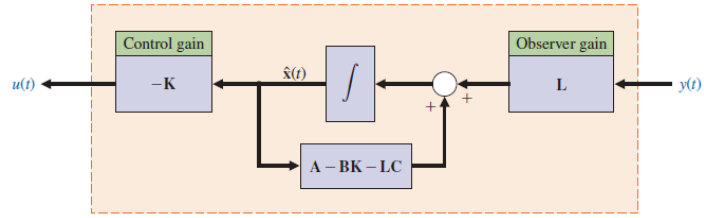


FIGURE 11.8
State variable compensator with integrated full-state feedback and observer.

O sa rpesuunem ca $e = x - \hat{x} \Rightarrow \dot{e} = \dot{x} - \dot{\hat{x}} = \mathrm{Ax} + \mathrm{Bu} - A\hat{x} - \mathrm{Bu} - \mathrm{Ly} + \mathrm{LC}\hat{x} =$ $A(x - \hat{x}) - \mathrm{Ly} + \mathrm{LC}\hat{x} = A(x - \hat{x}) - \mathrm{LCx} + \mathrm{LC}\hat{x} = (A - \mathrm{LC})(x - \hat{x}) = (A - \mathrm{LC})e$. Pe scurt, $\dot{e} = (A - \mathrm{LC})e$. Doamne ajuta, eroarea nu depinde de intrare

Reluand modelul de stare liniar $\dot{x} = \mathrm{Ax} + \mathrm{Bu} = \mathrm{Ax} - \mathrm{BF}\hat{x}$ $(\mathrm{Stim\ ca}\ u = -F\hat{x})$. Mai stim ca $e = x - \hat{x} \Rightarrow \hat{x} = x - e$. De aici mai dezvoltam asa: $\dot{x} = \mathrm{Ax} - \mathrm{BF}(x - e) = \mathrm{Ax} - \mathrm{BFx} + \mathrm{BFe} = (A - \mathrm{BF})x + \mathrm{BFe}$

Reamintim ca $\dot{e} = (A - \mathrm{LC})e$. Daca scriem $\begin{bmatrix} \dot{x} \\ \dot{e} \end{bmatrix} = \begin{bmatrix} A - \mathrm{BF} & \mathrm{BF} \\ 0 & A - \mathrm{LC} \end{bmatrix} \begin{bmatrix} x \\ e \end{bmatrix}$

Daca calculam valorile proprii ale sistemului in bucla inchisa, o sa fie reuniunea $\Lambda(A - \mathrm{BF}) \bigcup \Lambda(A - \mathrm{LC})$